

Sheet: VeraModule

VERA FPGA

File: vera-fpga.sch

Sheet: BusDecoder

BUS DECODER

File: busdecoder.sch

Sheet: Vera FPGA flash

VERA SPI FLASH
SD CARD INTERFACE

File: vera-fpga-flash.sch

Sheet: CartridgeInterface

CARTRIDGE
INTERFACE

File: cartridgeInterface.sch

Sheet: PowerSupply

POWER
SUPPLY

File: powersupply.sch

Sheet: VGA Analog

VGA
ANALOG

File: vga-analog.sch

FIDUCIAL TOP



FIDUCIAL BOTTOM



LOGO1
VERA X16 LOGO

LOGO2
ATARI READY

LOGO3
KICAD DESIGN

LOGO4
ATARI DUO BUS

LOGO5
ESP32_BOBOARD

LBL1

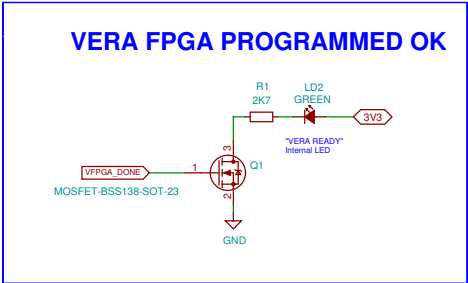
-  H1
MountingHole
-  H2
MountingHole
-  H3
MountingHole
-  H4
MountingHole

Gianluca Renzi
RetroBit Lab

Sheet: /
File: VERA-MODULE-RBL.sch

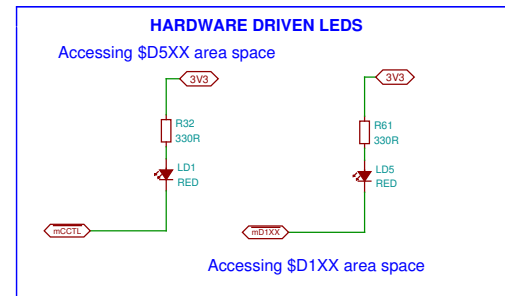
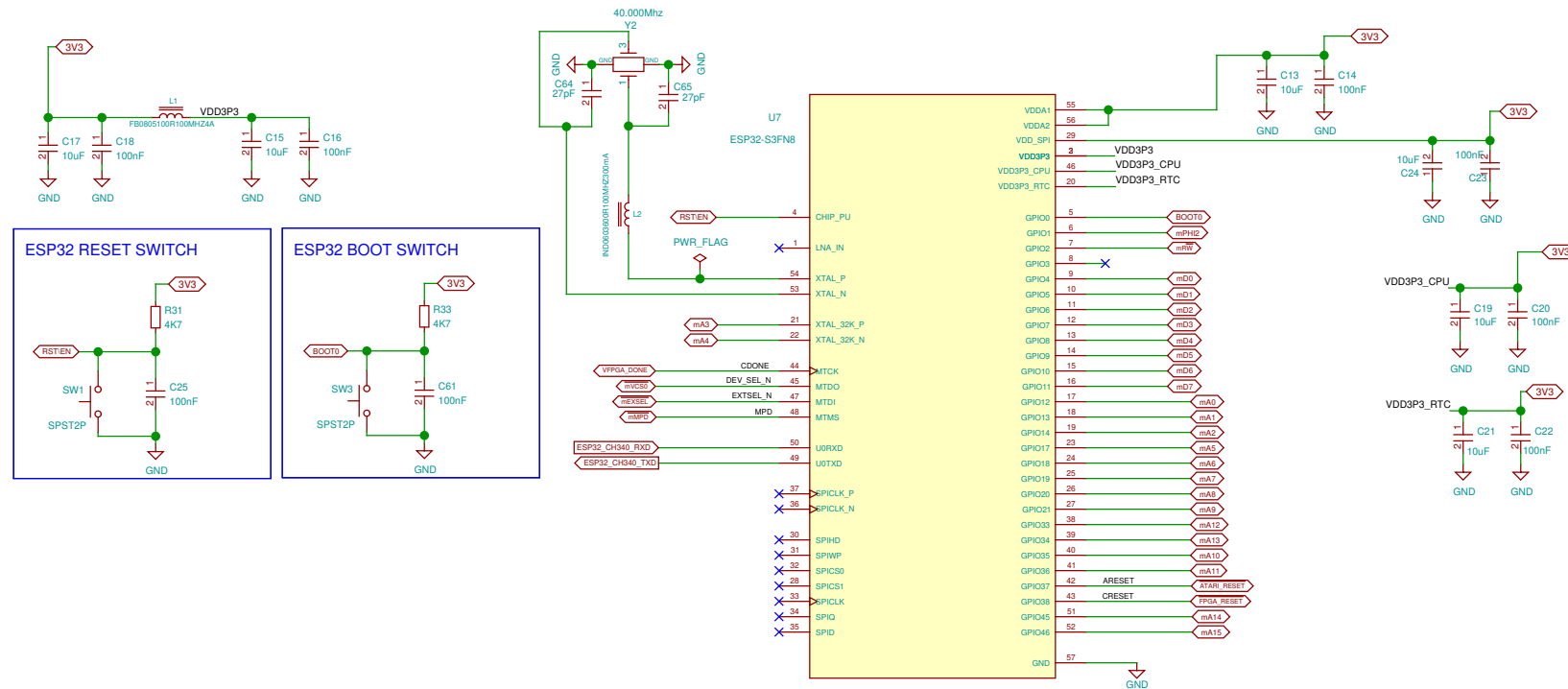
Title: VERA FPGA Audio & Video Board

Size: A4	Date: 2026-09-01	Rev: 1.0
Plotted by eeshow 3a954ac+ 20260903-06:45Z		Id: 1/7

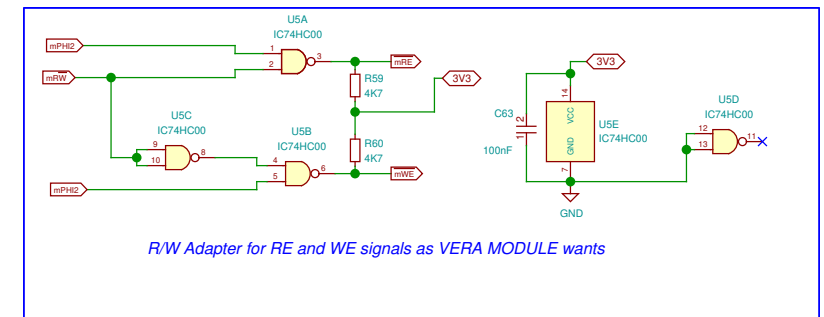


Title: VeraModule		
Size: A4	Date: 2026-09-01	Rev: 1.0
Plotted by eeshow 3a954ac+ 20260903-06:45Z		Id: 2/7

PBI DEVICE ID: software selectable only



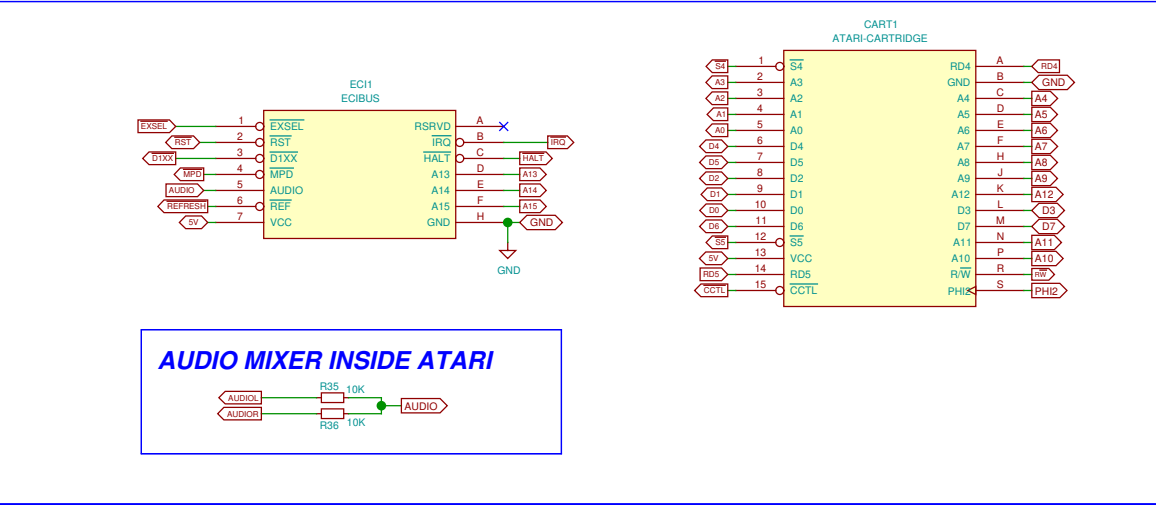
PBI Bus Interface Decoder:
I/O Space, Math Pack Disable, ROM ADDRESS (\$D800-\$DFFF),
RAM PBI SPACE (\$D600-\$D7FF)

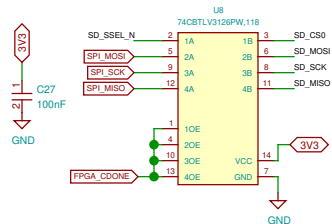


When VERA Chip Select (mVCS0) latch is active & A15..A0 \$D800-\$DFFF -> MPD active (External 2K ROM)
 When VERA Chip Select (mVCS0) latch is active & A15..A0 \$D600-\$D7FF -> EXSEL active (External 512 bytes RAM)
 When access @ \$D1xx and VERA Chip Select (mVCS0) latch is active -> EXSEL active

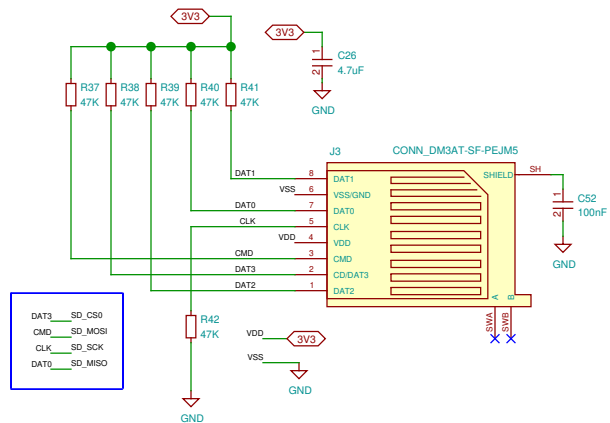
ATARI 130XE ECI & CARTRIDGE INTERFACE

All 8-Bit signals must be shifted from 5V to 3.3V and vice versa



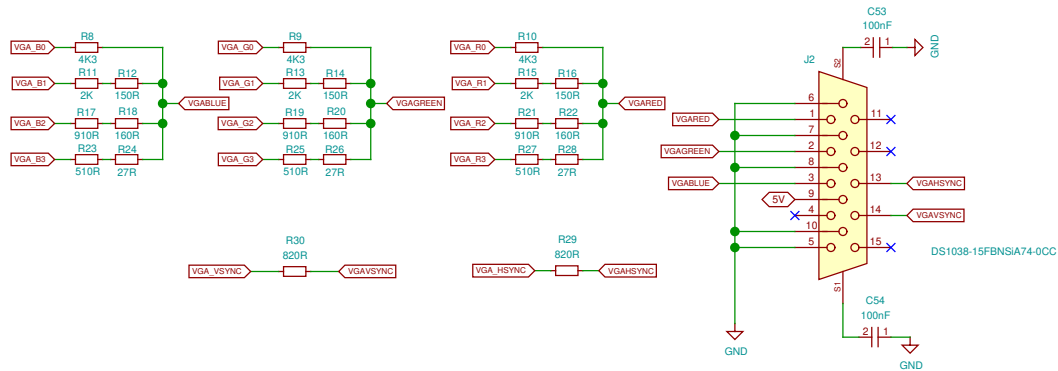


SD/microSD INTERFACE

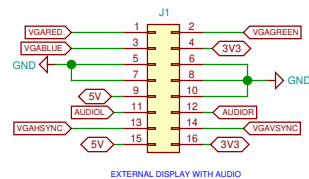


Rev: 1.0
Id: 5/7

ANALOG VGA SIGNALS



EXTERNAL VIDEO CONNECTOR



Gianluca Renzi

RetroBit Lab

Sheet: /VGA Analog/

File: vga-analog.sch

Title: VGA Analog

Size: A4

Date: 2026-09-01

Rev: 1.0

Plotted by eeshow 3a954ac+ 20260903-06:45Z

Id: 7/7